

ADS Assignments for Lab Examination

Div: SY-CSE (AI & ML)-D

AY 2025-26 (Sem-2)

1. Write a C program to implement Threaded Binary Tree and perform an Inorder traversal
2. Write a C program to implement Binary Search Tree Deletion.
 - Implement all 3 cases of deletion.
 - Display the resultant tree after the deletion of nodes
 - Perform Inorder, Preorder and Postorder traversals
3. Write a C program to implement Threaded Binary Search Tree and perform an Inorder traversal.
4. Write a C program to implement AVL Tree and perform following operations:
 - Insert node
 - Search node
 - Delete Node
 - Display tree using inorder traversal
5. Write a C program to implement Heap Sort using Max Heap Tree Concept
6. Write a C program to implement B-Tree and perform following operations:
 - Insert a node
 - Display tree by Inorder Traversal
7. Write a C program to implement Priority Queue for Job Queue using Max and Min Heap concept.
 - Job queue should consist of Job No., Job Name, Job Priority
 - Display the entire Job Queue
 - Pick the Job with highest priority and print the details of it.
 - Pick the Job with lowest priority and print the details of it.

8. Write a C program to implement a Red Black Tree. And perform the following operations:
 - Insert a node
 - Display tree
 - Search a node
9. Write a C program to implement a Skip List. And perform the following operations:
 - Insert a node
 - Create lists of level 0 to n-1
 - Display skip list
 - Search a node
10. Write a C program to implement a Trie (Prefix Tree) and perform following operations:
 - Insert a word in trie
 - Search a word in trie
 - Check if prefix exists
11. Write a C program to implement a Treap tree and perform the following operations:
 - Insert a key in treap
 - Search a key in treap
 - Delete a key in treap
 - Perform Inorder Traversal
12. Write a C program to implement a Segment tree and perform following operations:
 - Build a tree
 - Update value at an index
 - Print the sum
 - Range Query

Subject Faculty :
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